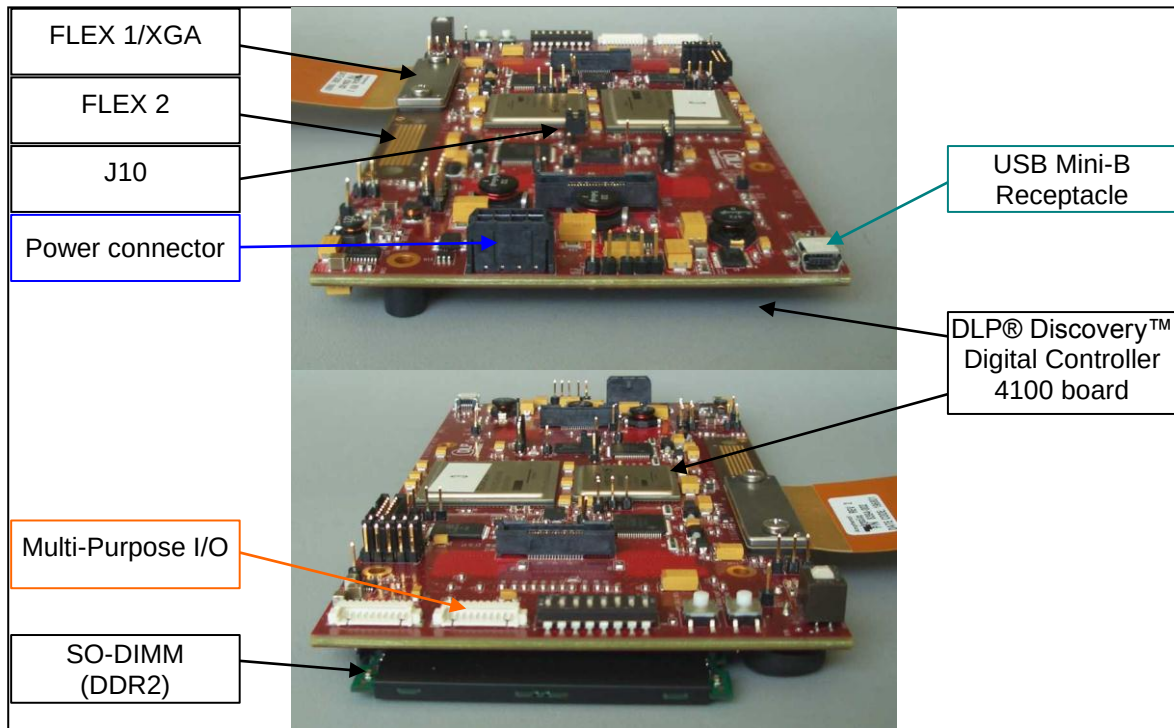


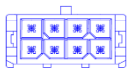
External Connections and Ports

The ALP-4.1 *high-speed* hardware consists of the DLP® Discovery™ 4100 Developer Kit including a memory module and a battery supported FPGA key. This document provides details about the electrical interfaces.



Connectors

On the DLP® Discovery™ Board, an 8-pin Molex DC power jack is provided to bring power in from a power supply unit.



Power connector (Molex Micro-Fit 3.0™ Header, 8 Circuits)

This is a Molex micro fit 3.0 dual row header, part number 43045-0812. It mates with Micro-Fit 3.0™ receptacle 43025-08xx.

Pin Number	Signal Name	Power Consumption
1 – 4 (lower on drawing)	GND	-
5 – 8	5 V	6 A



USB Mini-B connector

This is the connector for USB connection to the PC. Use the supplied USB cable to connect the board.

Pin Number	Signal Name
1 (left on photo)	USB VCC
2	USB D–
3	USB D+
4	not connected
5	GND



Multi-Purpose I/O (Synchronization) connector (Molex PicoBlade™ Header, 10 Circuits)

The synchronization connector of ALP-4.1 *high-speed* is connector J6 on the DLP® Discovery™ 4100 Developer Kit board. It is a Molex header, part number 53261-1071. It mates with Molex part number 51021-1000. Use crimp contacts 50079-8000.

This connector is not hot-plug capable. Only attach a plug while the device is off in order to avoid damage. Always apply a common GND when connecting devices.

Two pins are connected to the 2.5 V power regulator and ground. The other eight pins are logic input and output signals with 2.5 V LVCMOS IO standard. Input and In/Out pins have weak pull-ups. For details please consult the documentation of the Xilinx Virtex-5 FPGA: www.xilinx.com/support/documentation/data_sheets/ds202.pdf (Table 7: SelectIO DC Input and Output Levels and Table 3: DC Characteristics Over Recommended Operating Conditions). Drive strength is 2 mA.

Pin Number	Signal Name	Direction	Purpose
1 (left on photo)	2.5 V	—	Power
2	SYNCH_OUT1_GATE	OUT/Z	Dynamic Gate Synch Output 1, refer to the ALP-4 <i>high-speed</i> API description, ALP_DEV_DYN_SYNCH_OUT1_GATE
3	SYNCH_OUT2_GATE	OUT/Z	ALP_DEV_DYN_SYNCH_OUT2_GATE
4	SYNCH_OUT3_GATE	OUT/Z	ALP_DEV_DYN_SYNCH_OUT3_GATE
5	SPARE4	IN/OUT	
6	SPARE5	IN/OUT	
7	TRIGGER_IN	IN	Trigger Input
8	SYNCH_OUT	OUT	Frame Synchronization Output

Pin Number	Signal Name	Direction	Purpose
9	PWM	OUT	ALP_PWM_LEVEL: output a pulse-width modulated value
10	GND	—	Power

WARNING: This connector is directly connected to the FPGA. There is no over-voltage protection, debounce, nor similar circuitry in between.

NOTE: Higher input voltage is allowed only in conjunction with a series resistance. For 5 V we recommend a resistor of 2.2 kΩ. If deeper understanding is required, please consult the Virtex-5 Data Sheet (see above) and the Virtex-5 User Guide:

www.xilinx.com/support/documentation/user_guides/ug190.pdf

(Section 3.3V I/O Design Guidelines).

SO-DIMM Socket

A standard DDR2 SO-DIMM (small outline dual-inline memory module) is supplied with ALP-4.1 *high-speed*. Please always use the module specified and supplied by ViALUX. Always switch off the device before removal or insertion of the memory module.

FLEX 1/XGA, FLEX 2, and J10

When using an XGA DMD please connect FLEX 1 and open jumper J10. The 1080p DMD requires J10 being installed and both flex cables connected.

Always switch the device off before exchanging the DMD.

Mounting Instructions and Measures

Please refer to drawings supplied in the DLP® Discovery™ Knowledge Base for dimensions of the DLP® Discovery™ 4100 Developer Kit

Never drill into a multi-layer board!

Please be aware that there is an on-board battery that supplies memory for FPGA encryption. Avoid short-circuits when mounting the board in order to not lose the encryption key.

The ALP system is designed to work in any mounting position. Good air circulation is recommended.